

# fixed

## Fixed

**Revision:** 29 **Last modified:** 2026-06-01T21:56:42Z **Description:** Completed workable items (with evidence references) **Authority:** Constitution §11.4.93 (workable-items DB single source of truth) **Generated-by:** scripts/docs/db\_to\_md.py (DB is canonical; edit via cmd/hxc-registry, not by hand)

Total completed: **180**.

| HXC      | Type    | Pri | Title                                                                     | Commit  |
|----------|---------|-----|---------------------------------------------------------------------------|---------|
| HXC-001  | Task    | P1  |                                                                           |         |
| HXC-002  | Task    | P1  |                                                                           |         |
| HXC-003  | Task    | P1  |                                                                           |         |
| HXC-004  | Task    | P1  |                                                                           |         |
| HXC-005  | Task    | P1  |                                                                           |         |
| HXC-006  | Task    | P1  |                                                                           |         |
| HXC-007  | Task    | P1  |                                                                           |         |
| HXC-008  | Task    | P1  |                                                                           |         |
| HXC-009  | Task    | P1  |                                                                           |         |
| HXC-010  | Task    | P1  |                                                                           |         |
| HXC-011  | Task    | P1  |                                                                           |         |
| HXC-012  | Task    | P1  |                                                                           |         |
| HXC-013  | Task    | P1  |                                                                           |         |
| HXC-014  | Task    | P1  |                                                                           |         |
| HXC-015  | Task    | P1  |                                                                           |         |
| HXC-1001 | Task    | P0  | Stand up embedded-etcd integration harness for pkg/etcd tests             | dbed10c |
| HXC-1002 | Feature | P0  | Prove etcd Put/Get round-trip and missing-key semantics against real etcd | dbed10c |
| HXC-1003 | Feature | P1  | Prove etcd GetPrefix returns exactly matching multi-key set               | dbed10c |
| HXC-1004 | Feature | P1  | Prove etcd Delete/DeletePrefix remove keys (sink-side absence)            | dbed10c |
| HXC-1005 | Feature | P0  | Prove etcd Watch fires correct event and closes on ctx cancel             | dbed10c |

| HXC      | Type    | Pri | Title                                                                                             | Commit              |
|----------|---------|-----|---------------------------------------------------------------------------------------------------|---------------------|
| HXC-1006 | Feature | P1  | Prove etcd Lease/PutWithLease/RevokeLease/KeepAlive TTL behavior                                  | dbed10c             |
| HXC-1007 | Feature | P0  | Prove etcd distributed Lock mutual exclusion (highest-risk path)                                  | dbed10c             |
| HXC-1008 | Task    | P2  | Add mutation-paired etcd New() default-timeout injection test                                     | dbed10c             |
| HXC-1009 | Task    | P2  | Prove etcd error-wrapping on dead endpoint                                                        | dbed10c             |
| HXC-1010 | Feature | P0  | Replace pkg/infra map-echo orchestrator with real service Boot via testcontainers                 | 85ee85e85f221075dfc |
| HXC-1011 | Feature | P0  | Make pkg/infra Health probe real services and report false on failure                             | 85ee85e85f221075dfc |
| HXC-1012 | Feature | P1  | Make pkg/infra Logs capture real sink-side output honoring tail/since/timestamps                  | 85ee85e85f221075dfc |
| HXC-1013 | Feature | P1  | Make pkg/infra Scale create real replicas verified via runtime query                              | 85ee85e85f221075dfc |
| HXC-1014 | Feature | P1  | Make pkg/infra VMspawn/VMSSH establish a real session end-to-end                                  | 44db9182bb5abe8f00f |
| HXC-1015 | Feature | P2  | Make pkg/infra VMSimulateFailure/VMSimulatePartition assert observable network effect             | 85ee85e85f221075dfc |
| HXC-1016 | Feature | P0  | Add EtcdLocker integration tests against real etcd (block/acquire/lease)                          | dbed10c             |
| HXC-1017 | Bug     | P0  | Rewrite TestMemoryLockerConcurrent to detect a broken lock without -race                          | dbed10c             |
| HXC-1018 | Task    | P2  | Add explicit block-then-acquire ordering test for MemoryLocker                                    | dbed10c             |
| HXC-1019 | Bug     | P2  | Characterize or fix MemoryLocker.Lock context-cancellation goroutine leak                         | 32feec3             |
| HXC-1020 | Feature | P0  | Wire a real Builder behind pkg/build Service and prove image is pullable                          | 85ee85e85f221075dfc |
| HXC-1021 | Bug     | P1  | Drive pkg/build StateFailed via a genuine build failure not the 'fail' sentinel                   | dbed10c             |
| HXC-1022 | Bug     | P1  | Make pkg/build List/Concurrent tests Start() the service and assert terminal states               | 32feec3             |
| HXC-1023 | Task    | P2  | Replace time.Sleep polling in pkg/build with completion sync and add state-machine mutation tests | 32feec3             |
| HXC-1024 | Bug     | P2  | Enforce or document pkg/build content-addressable cache digest integrity                          | dbed10c             |
| HXC-1025 | Task    | P2  | Cover pkg/build cancel-mid-flight branch sets StateCancelled                                      | 32feec3             |
| HXC-1026 | Feature | P0  | Add real mTLS handshake test for pkg/security (valid client succeeds, no/foreign cert rejected)   | dbed10c             |
| HXC-1027 | Bug     | P1  | Assert MinVersion==TLS13 on the client TLS path in pkg/security                                   | dbed10c             |

| HXC      | Type    | Pri | Title                                                                                | Commit              |
|----------|---------|-----|--------------------------------------------------------------------------------------|---------------------|
| HXC-1028 | Feature | P0  | Locate/build production Vault client and add dev-Vault integration test (KVv2 + PKI) | 85ee85e85f221075dfc |
| HXC-1029 | Task    | P3  | Replace pkg/security non-nil-only TestNewTLSConfigBuilder with meaningful assertion  | 32feec3             |
| HXC-1030 | Bug     | P1  | Verify pkg/hxcregistry item_history audit-sink rows after Create/Update              | dbed10c             |
| HXC-1031 | Bug     | P2  | Assert exact and stable ComputeHeadingHash value in pkg/hxcregistry                  | dbed10c             |
| HXC-1032 | Task    | P2  | Add pkg/hxcregistry failure-path tests (dup PK, missing-id Get/Update)               | dbed10c             |
| HXC-1033 | Bug     | P2  | Make pkg/hxcregistry concurrent test do concurrent writes (or rename)                | dbed10c             |
| HXC-1034 | Task    | P3  | Prove pkg/hxcregistry timestamp round-trip and DB CHECK-constraint enforcement       | dbed10c             |
| HXC-1035 | Bug     | P0  | Wire pkg/wasm WasmPlugin.Execute to real WASM output and prove transformed result    | dbed10c             |
| HXC-1036 | Task    | P1  | Add pkg/wasm negative tests for missing execute export and corrupt module Init       | dbed10c             |
| HXC-1037 | Bug     | P2  | Prove pkg/wasm init/shutdown exports actually run via memory side-effect             | dbed10c             |
| HXC-1038 | Task    | P3  | Cover pkg/wasm Host.Call ErrUnexpectedResult branch                                  | 32feec3             |
| HXC-1039 | Bug     | P1  | Replace self-defeating pkg/semaphore TestSemaphore_ReleaseTooMany_Mutation           | a084eb1             |
| HXC-1040 | Feature | P1  | Add pkg/semaphore M>>N concurrency invariant test under -race                        | a084eb1             |
| HXC-1041 | Task    | P3  | Add pkg/semaphore New(0) barrier and exact-capacity boundary tests                   | a084eb1             |
| HXC-1042 | Bug     | P1  | Convert pkg/storage error tests to errors.Is on exported sentinels                   | dbed10c             |
| HXC-1043 | Bug     | P2  | Reconcile and test pkg/storage FileStore vs MemoryStore copy semantics               | dbed10c             |
| HXC-1044 | Task    | P2  | Add pkg/storage overwrite, missing-delete, and List edge tests for both stores       | dbed10c             |
| HXC-1045 | Feature | P1  | Add pkg/storage -race concurrent Put/Get/Delete test to prove RWMutex                | dbed10c             |
| HXC-1046 | Bug     | P1  | Gate -race in CI for pkg/metrics concurrency mutation tests                          | a084eb1             |
| HXC-1047 | Bug     | P2  | Complete pkg/metrics Prometheus exposition assertions (+Inf, _sum, ordering)         | a084eb1             |
| HXC-1048 | Task    | P2  | Add pkg/metrics duplicate-registration and histogram edge tests                      | a084eb1             |
| HXC-1049 | Task    | P3  | Add pkg/metrics /metrics integration test via httptest.Server                        | a084eb1             |

| HXC      | Type    | Pri | Title                                                                                       | Commit  |
|----------|---------|-----|---------------------------------------------------------------------------------------------|---------|
| HXC-1050 | Bug     | P1  | Test pkg/grpcutil LoggingStreamInterceptor recv/send counters                               | a084eb1 |
| HXC-1051 | Bug     | P2  | Strengthen pkg/grpcutil log assertions to dynamic behavior-dependent fields                 | a084eb1 |
| HXC-1052 | Task    | P2  | Cover pkg/grpcutil status.FromError code extraction for handler errors                      | a084eb1 |
| HXC-1053 | Bug     | P1  | Add pkg/context TestDetach_PreservesValues for the value-retention guarantee                | a084eb1 |
| HXC-1054 | Bug     | P2  | Strengthen pkg/context WithTimeout deadline and expiry assertions                           | a084eb1 |
| HXC-1055 | Bug     | P1  | Give pkg/workerpool TestPool_NilWorkSafe_Mutation a real sink                               | a084eb1 |
| HXC-1056 | Feature | P1  | Add pkg/workerpool fan-in count==N, post-Stop Submit, panic-recovery, idempotent-Stop tests | a084eb1 |
| HXC-1057 | Bug     | P1  | Make pkg/errors concurrency test honest via locked GetFields path + -race                   | a084eb1 |
| HXC-1058 | Task    | P2  | Pin pkg/errors Code enum exact values, distinctness, and WithFields nil receiver            | a084eb1 |
| HXC-1059 | Bug     | P2  | Replace tautological pkg/backoff cap-mutation test and pin Default()                        | a084eb1 |
| HXC-1060 | Bug     | P2  | Assert pkg/classads parser AST Op per operator and paren precedence                         | a084eb1 |
| HXC-1061 | Task    | P3  | Add pkg/classads eval failure-path and regexp non-match tests                               | a084eb1 |
| HXC-1062 | Bug     | P2  | Add pkg/crypto SHA-256 known-answer vectors and sentinel errors.Is checks                   | dbed10c |
| HXC-1063 | Task    | P3  | Add pkg/crypto AES-128/192, wrong-key decrypt, short-ciphertext, PBKDF2 vector tests        | dbed10c |
| HXC-1064 | Bug     | P1  | Add pkg/log prefix-field proof catching the SetLevel prefix-loss bug                        | a084eb1 |
| HXC-1065 | Task    | P3  | Strengthen pkg/log Warn test, add Debug-suppression, remove dead captureOutput scaffolding  | a084eb1 |
| HXC-1066 | Task    | P3  | Add pkg/lru boundary, cold-miss, and struct-value round-trip tests                          | a084eb1 |
| HXC-1067 | Feature | P2  | Add pkg/pubsub -race concurrency test and explicit drop-boundary assertion                  | a084eb1 |
| HXC-1068 | Bug     | P2  | Prove pkg/ratelimit bucket starts full and add Limiter conformance + -race tests            | a084eb1 |
| HXC-1069 | Bug     | P1  | Bring pkg/retry DoWithResult to behavioral parity with Do                                   | a084eb1 |
| HXC-1070 | Task    | P2  | Add pkg/retry transient-recovery, jitter-bound, and computeDelay table tests                | a084eb1 |
| HXC-1071 | Task    | P3  |                                                                                             | a084eb1 |

| HXC      | Type    | Pri | Title                                                                             | Commit              |
|----------|---------|-----|-----------------------------------------------------------------------------------|---------------------|
|          |         |     | Strengthen pkg/serde MustMarshal exact-content and add Unmarshal failure tests    |                     |
| HXC-1072 | Bug     | P2  | Pin pkg/validator error contract via errors.Is and remove/wire unused sentinels   | a084eb1             |
| HXC-1073 | Task    | P3  | Add pkg/validator uint min/max, IsValidID empty, oneof/email boundary tests       | a084eb1             |
| HXC-1074 | Task    | P0  | Establish fleet-wide -race CI gate across concurrency-sensitive packages          | a084eb1             |
| HXC-1075 | Feature | P0  | Implement NATS/JetStream client wrapper (pkg/events)                              | 44db9182bb5abe8f00f |
| HXC-1076 | Feature | P0  | Implement etcd client wrapper (pkg/etcd)                                          | 44db9182bb5abe8f00f |
| HXC-1077 | Feature | P0  | Implement etcd discovery backend (pkg/discovery)                                  | 44db9182bb5abe8f00f |
| HXC-1078 | Feature | P0  | Implement real Kafka producer/consumer (pkg/pubsub)                               | 44db9182bb5abe8f00f |
| HXC-1079 | Feature | P0  | Implement cgroups v2 + /proc resource reader (pkg/resources)                      | 44db9182bb5abe8f00f |
| HXC-1080 | Feature | P0  | Implement Omega-model scheduler core (pkg/scheduler, internal/scheduler)          | 44db9182bb5abe8f00f |
| HXC-1081 | Feature | P0  | Implement optimistic-concurrency versioning for scheduler pool                    | 44db9182bb5abe8f00f |
| HXC-1082 | Feature | P1  | Implement HTCondor ClassAds matching in scheduler (pkg/classads)                  | 44db9182bb5abe8f00f |
| HXC-1083 | Feature | P1  | Implement gang scheduling and preemption in scheduler                             | 44db9182bb5abe8f00f |
| HXC-1084 | Feature | P0  | Upgrade tmux backend to control-mode (-CC) with streaming I/O (pkg/session)       | 1367dca3bd532a491ac |
| HXC-1085 | Feature | P0  | Implement CRDT session state and lifecycle (create/attach/migrate)                | 1367dca3bd532a491ac |
| HXC-1086 | Feature | P0  | Implement real Node gRPC handlers serving helixv1 Node API (internal/node)        | 4d0ac6681bf586c30aa |
| HXC-1087 | Feature | P0  | Make node registry etcd-backed and cross-process shared (internal/node)           | 2f27ac9001d3fc0a900 |
| HXC-1088 | Feature | P0  | Implement SWIM gossip membership and failure detection (pkg/swim)                 | 4d0ac6681bf586c30aa |
| HXC-1089 | Feature | P0  | Implement WireGuard mesh config-gen and key rotation (pkg/wireguard)              | 2f27ac9001d3fc0a900 |
| HXC-1090 | Feature | P1  | Implement WireGuard NAT traversal / STUN hole-punch (pkg/wireguard)               | 1367dca3bd532a491ac |
| HXC-1091 | Feature | P0  | Wire API gateway reverse proxy with prefix routing and /health (internal/gateway) | 2f27ac9001d3fc0a900 |
| HXC-1092 | Feature | P0  | Add JWT/RBAC auth middleware to API gateway (internal/gateway)                    | 2f27ac9001d3fc0a900 |
| HXC-1093 | Feature | P0  | Implement leader election with fencing tokens (pkg/leader)                        | 4d0ac6681bf586c30aa |

| HXC      | Type    | Pri | Title                                                                            | Commit               |
|----------|---------|-----|----------------------------------------------------------------------------------|----------------------|
| HXC-1094 | Feature | P0  | Implement distributed lock primitives over etcd (pkg/lock)                       | 2f27ac9001d3fc0a900  |
| HXC-1095 | Feature | P1  | Implement Postgres-backed internal artifact registry (pkg/hxcregistry)           | 1367dca3bd532a491ac  |
| HXC-1096 | Feature | P1  | Implement GPU resource probe with /proc parser (internal/gpu)                    | 9c58146              |
| HXC-1097 | Feature | P1  | Define vendor-agnostic GPUBackend interface and registry (internal/gpu)          | 05f408d              |
| HXC-1098 | Feature | P2  | Implement GPU sharing modes (MPS, time-slice, MIG, exclusive)                    | 1a67f0a              |
| HXC-1099 | Feature | P1  | Implement Build Service orchestrator and worker pool (internal/build)            | 1367dca3bd532a491ac  |
| HXC-1101 | Feature | P0  | Implement e2ee transport record protocol (security/pkg/e2ee)                     | e12003abbbb68db0e08f |
| HXC-1102 | Feature | P1  | Implement software-rooted attestation (security/pkg/attestation)                 | e12003abbbb68db0e08f |
| HXC-1103 | Feature | P0  | Implement RBAC scopes and identity bindings (internal/security)                  | e12003abbbb68db0e08f |
| HXC-1104 | Feature | P1  | Implement SPIFFE/SPIRE SVID issuance and mTLS identity (internal/security)       | 7634fbbfaf4e66eba233 |
| HXC-1105 | Feature | P0  | Implement htmux CLI raw-mode terminal client (cmd/htmux)                         | e12003abbbb68db0e08f |
| HXC-1106 | Feature | P0  | Implement session I/O PTY-over-WebSocket forwarding (cmd/helix-session)          | e12003abbbb68db0e08f |
| HXC-1107 | Feature | P0  | Implement WebSocket envelope and MessagePack framing (pkg/websocket)             | e12003abbbb68db0e08f |
| HXC-1108 | Feature | P1  | Implement HelixQA live-service Challenge runner (cmd/helix-test)                 | 6c825dfc56a498b3715  |
| HXC-1109 | Feature | P1  | Implement deterministic-sim, chaos, and device-sim test harness (cmd/helix-test) | 6c825dfc56a498b3715  |
| HXC-1110 | Task    | P1  | Wire Prometheus /metrics endpoint uniformly across all services (cmd/helix-*)    | 1a67f0a              |
| HXC-1111 | Task    | P1  | Wire Jaeger/OTel trace-ID propagation across services (pkg/tracing)              | 05f408d              |
| HXC-1112 | Task    | P1  | Wire pkg/health gRPC health protocol on every service (pkg/health, cmd/helix-*)  | 05f408d              |
| HXC-1114 | Feature | P0  | Provision PostgreSQL primary schema with 15 tables, indexes, and triggers        | e12003abbbb68db0e08f |
| HXC-1115 | Task    | P0  | Define etcd key namespace and key-builder constants (pkg/etcd)                   | e12003abbbb68db0e08f |
| HXC-1116 | Feature | P1  | Define Redis cache, routing, and rate-limit key structure (pkg/storage)          | 7634fbbfaf4e66eba233 |
| HXC-1118 | Feature | P1  | Implement REST API surface on API Gateway with OpenAPI 3.0                       | 6c825dfc56a498b3715  |
| HXC-1120 | Task    | P1  |                                                                                  | 6c825dfc56a498b3715  |

| HXC      | Type    | Pri | Title                                                                                  | Commit               |
|----------|---------|-----|----------------------------------------------------------------------------------------|----------------------|
|          |         |     | Configure Kafka topics with partitions/<br>replication/retention                       |                      |
| HXC-1121 | Task    | P0  | Configure NATS JetStream streams<br>(HELIX_NODES/SESSIONS/SCHEDULER/<br>HEALTH/ALERTS) | 6c825dfc56a498b3715  |
| HXC-1122 | Feature | P2  | Implement Cap'n Proto zero-copy serialization<br>for control-plane (pkg/serde)         | 63822b3bcd439a360fa  |
| HXC-1123 | Feature | P2  | Implement FlatBuffers serialization for GPU<br>compute payloads                        | 63822b3bcd439a360fa  |
| HXC-1124 | Feature | P2  | Implement ZeroMQ message patterns for<br>internal data planes                          | 63822b3bcd439a360fa  |
| HXC-1125 | Feature | P1  | Implement Setup Wizard single-command node<br>onboarding (cmd, BASH+Go)                | 9c58146              |
| HXC-1127 | Feature | P1  | Implement OPA/WASM Policy Engine with<br>HelixConstitution enforcement                 | 7634fbbfaf4e66eba233 |
| HXC-1131 | Feature | P2  | Implement Backup Service (etcd snapshots,<br>WAL archival, Ceph checkpoints)           | 467a6dc62cc792f48a6  |
| HXC-1132 | Feature | P1  | Implement Metrics Collector node scrape<br>endpoint and GPU metrics aggregation        | 7634fbbfaf4e66eba233 |
| HXC-1133 | Feature | P1  | Implement token-bucket rate limiter for API<br>gateway (pkg/ratelimit)                 | e12003abbb68db0e08f  |
| HXC-1134 | Feature | P2  | Implement Vault secret injection and rotation for<br>all services                      | 467a6dc62cc792f48a6  |
| HXC-1135 | Feature | P0  | Implement two-node cluster formation end-to-<br>end (exit gate)                        | 1c0d943              |
| HXC-1136 | Feature | P0  | Implement session-create end-to-end path<br>(htmux new exit gate)                      | 553a043              |
| HXC-1137 | Task    | P1  | Benchmark job scheduling decision latency<br>under 100ms (exit gate)                   | 553a043              |
| HXC-1138 | Task    | P1  | Enforce >60% pkg/ line coverage and paired<br>mutation tests (Constitution 1.1)        | 75f076f              |
| HXC-1141 | Feature | P2  | Implement interactive-mode AI CLI agent<br>resource provisioning                       | 1a67f0a              |
| HXC-1142 | Feature | P2  | Implement hybrid mode switching and batch/<br>interactive resource sharing             | 9c58146              |
| HXC-1144 | Feature | P2  | Implement WireGuard mesh network policies<br>(segmentation)                            | 467a6dc62cc792f48a6  |
| HXC-1148 | Feature | P1  | Replace pkg/wireguard NAT discovery with<br>RFC 5389 STUN client                       | 7718c20              |
| HXC-1149 | Task    | P3  | Return typed ErrUnsupported for UPnP/NAT-<br>PMP in pkg/wireguard                      | 7718c20              |
| HXC-1150 | Feature | P1  | Add WireGuard key-rotation grace/overlap<br>window with ActiveKeys()                   | 7718c20              |
| HXC-1151 | Task    | P1  | Add pkg/discovery etcd-backed integration test<br>against a real etcd                  | 7718c20              |

| HXC      | Type    | Pri | Title                                                                                    | Commit  |
|----------|---------|-----|------------------------------------------------------------------------------------------|---------|
| HXC-1156 | Feature | P1  | Implement thermal-aware scheduler plugin (throttle on overheat)                          | 7718c20 |
| HXC-1157 | Feature | P1  | Implement semi-trusted node output verification (redundant-compute/checksum)             | 7718c20 |
| HXC-1188 | Feature | P0  | Enforce work-unit resource limits (duration/memory/CPU) on edge devices                  | 7718c20 |
| HXC-1189 | Feature | P0  | Implement EdgeAwarePlugin scheduler Filter stage                                         | 203c42a |
| HXC-1190 | Feature | P1  | Implement EdgeAwarePlugin scheduler Score stage                                          | 203c42a |
| HXC-1191 | Feature | P1  | Implement declarative per-tier ScheduleRule engine                                       | 7718c20 |
| HXC-1195 | Feature | P0  | Implement edge trust-level model (STANDARD/SEMI/EDGE_DONOR)                              | 7718c20 |
| HXC-1196 | Feature | P0  | Enforce workload restriction matrix by trust level                                       | 7718c20 |
| HXC-1198 | Feature | P1  | Implement offline sync protocol with delta compression                                   | 203c42a |
| HXC-1199 | Feature | P1  | Implement edge sensor-fusion framework and stream workload type                          | 75f076f |
| HXC-1210 | Bug     | P0  | Wire EtcdBackend into the node agent for cluster-wide discovery                          | 203c42a |
| HXC-1211 | Bug     | P0  | Make internal/scheduler.StreamJobEvents a real event stream                              | 7718c20 |
| HXC-1212 | Feature | P0  | Connect pkg/classads to scheduler Filter/Score stages                                    | 7718c20 |
| HXC-1213 | Feature | P0  | Add auth + RBAC enforcement to the gateway                                               | 203c42a |
| HXC-1214 | Feature | P1  | Add S3-compatible (minio) backend to pkg/storage                                         | 7718c20 |
| HXC-1215 | Feature | P1  | Add OTLP/Jaeger export shim for pkg/tracing                                              | 7718c20 |
| HXC-1220 | Task    | P2  | Replace custom policy engine with OPA/Rego                                               | 203c42a |
| HXC-1229 | Task    | P0  | Build device profile registry with versioned YAML schema for T1-T8                       | 1c0d943 |
| HXC-1231 | Feature | P1  | Implement device provisioning lifecycle abstraction (Provisioner/Instance state machine) | 553a043 |
| HXC-1238 | Feature | P1  | Implement BUGGIFY macro framework with ~25% deterministic fire rate                      | 75f076f |
| HXC-1239 | Feature | P1  | Implement 10:1 DST virtual-time compression by fast-forwarding idle periods              | 75f076f |
| HXC-1240 | Feature | P1  | Achieve 1,000+ simulated nodes in a single DST process                                   | 75f076f |
| HXC-1241 | Feature | P1  | Author DST consensus+gossip workloads using SETUP→EXECUTION→CHECK→METRICS pattern        | 75f076f |
| HXC-1247 | Feature | P1  |                                                                                          | 1c0d943 |



| HXC      | Type    | Pri | Title                                                                                                   | Commit  |
|----------|---------|-----|---------------------------------------------------------------------------------------------------------|---------|
|          |         |     | Implement pure-Go in-sim chaos faults toward 25+ (ClockSkew, DiskFill, MessageReorder, Byzantine, etc.) |         |
| HXC-1260 | Feature | P1  | Implement metrics validation against baseline KPI table with severity gating                            | 553a043 |
| HXC-1261 | Feature | P1  | Implement Welch's t-test statistical regression detector for HelixQA                                    | 1c0d943 |
| HXC-902  | Task    | P1  |                                                                                                         |         |
| HXC-903  | Task    | P1  |                                                                                                         |         |